Form 2

The Patent Act 1970

(39 of 1970)

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The patent rules, 2003

Complete specification

(See section 10 and rule 13)

Title of the invention:

Autonomous Multi-Agent Hybrid Healthcare Intelligence System

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PREAMBLE TO THE DESCRIPTION

[0001] The disclosed embodiments herein particularly describe the invention and the manner in which it is to be designed, developed, simulated, and performed. The embodiments are generally related to autonomous multi-agent artificial intelligence systems for healthcare guidance, implementing hierarchical architecture comprising multimodal medical data ingestion, specialized AI processing through six autonomous agents, and application orchestration layers. The six autonomous multi-agent artificial intelligent systems include medicine guidance, disease analysis, clinical document interpretation, geolocation discovery, symptom inference, and emergency triage, which utilize asynchronous artificial intelligent processing through the application of the transformer neural network with mixture-of-experts routing to generate contextual medical responses in multilingual frameworks without human intervention. The invention overcomes significant healthcare accessibility challenges that include geographical distances, consultation costs, linguistic incompatibility, and the unavailability of medical practitioners, providing autonomous intelligent health guidance that is uninterrupted and unending to all populations through the application of artificial intelligence independent of synchronous medical expertise to access preliminary medical information.

BACKGROUND OF THE INVENTION

[0002] With the exponential rise in demand for healthcare services in geographically spread-out populations, there is a need to provide accessible medical information for initial health decisions in these populations. There is substantial medical knowledge available in various pharmaceutical data, medical taxonomies, and treatment guidelines in various healthcare systems, but interpreting this heterogenous data for end user consumption requires specialized clinical expertise, multilingual abilities, and constant availability, which is beyond the scope of traditional healthcare delivery systems, particularly in underserved rural and economically disadvantaged populations.

[0003] In order to overcome the constraints related to the accessibility of healthcare services, digital health platforms such as telemedicine and chatbot systems were developed to provide medical guidance through digital means. However the telemedicine platforms follow the synchronous constraints related to the availability of scheduled medical consultations with the physician, while the chatbot systems follow the decision tree approach with general answers without any conceptual knowledge, making it difficult to provide autonomous health care intelligence.

[0004] The digital Healthcare platforms that are currently being developed focus on the delivery of information rather than providing autonomous intelligent reasoning capability. The information is provided through scheduled medical consultations and is required to be navigated by

the user without any personal guidance. The user is required to consult medical professionals, which makes the user dependent on the constraints related to medical availability. The systems do not support the generation of content in regional Indian languages are spoken by millions of people in the country.

[0005] Existing artificial intelligence-based healthcare assistants do not use specialized multi-agent systems for specific medical domains. Monolithic language models are not able to maintain in-depth information in pharmaceutical science, pathophysiology, emergency medicine, and diagnostic reasoning simultaneously and maintain domain specific expertise. No such system has implemented autonomous interpretation of clinical documents using optical character recognition and artificial intelligence-based analysis, probabilistic inference of symptoms and generation of differential diagnoses, and emergency classification using standardized risk stratification.

[0006] Existing prior art has shown critical deficiencies in the absence of mixture of exports routing for resource allocation to specific medical domains, lack of age-specific pharmaceutical guidance systems, absence of geolocation-based discovery of healthcare facilities and integration with intelligent guidance systems, lack of consultation history and exportable documentation systems, and insufficient native language generation systems maintaining precision in terminology. All such technological

deficiencies in prior art prevent the deployment of autonomous healthcare intelligence at the population level and necessitate the need for novel multi-agent artificial intelligence systems implementing specialized reasoning in the absence of human intervention.

SUMMARY OF THE INVENTION

[0007] The following summary is provided to facilitate an understanding of the innovative features unique to the disclosed embodiment. A full appreciation of the various aspects of the embodiments disclosed herein can be gained by taking the entire specification, claims, drawings, and abstract as a whole

[0008] It is, therefore, one aspect of the disclosed embodiments to provide an autonomous multi-agent artificial intelligence system for healthcare guidance utilizing hierarchical three-tier architecture consisting of multimodal medical data ingestion with validation protocols, specialized AI processing utilizing six autonomous agents with transformer-based neural networks and mixture-of-experts routing, and application orchestration providing contextualized medical responses across twelve linguistic frameworks. The system removes critical accessibility issues by providing continuous operation without synchronous medical practitioner dependency, providing preliminary healthcare guidance across geographical, economic, and linguistic boundaries. The invention democratizes healthcare information through the coordination of

autonomous agents for pharmaceutical guidance, disease analysis, clinical document interpretation, geolocation discovery, symptom inference, and emergency triage classification.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying figures, in which like reference numerals refer to identical or functionally similar elements throughout the separate views and which are incorporated in and form a part of the specification, further illustrate the present invention and, together with the detailed description of the invention, serve to explain the principles of the present invention.

[0010] FIG 1 Multi-Layer Intelligent Medical Assistance System Architecture

[0011] FIG 2 Illustrative Architecture of a Transformer-Based Language Model Including a Mixture-Of-Experts Processing Layer

DETAILED DESCRIPTION OF THE INVENTION

[0012] The particular details, values, and specifications presented in these non-limiting examples may be varied and are cited merely to illustrate at least one embodiment and are not intended to limit the present scope thereof.

[0013] The embodiments are described in detail with reference to the accompanying drawings, wherein illustrative implementations of the automated data analysis system are disclosed. The embodiments may be

realized in various forms and should not be construed as limited to the specific configurations described. Rather, the disclosure is intended to convey the full scope of the system to those skilled in the art. Like reference numerals refer to like elements throughout the drawings. As used herein, the term “and/or” includes any combination of one or more of the associated listed items.

[0014] **FIG. 1 illustrates** the Transformer-Based Language Model with Mixture of Exports Processing Layer, which includes tokenization, position embedding, dual RMS normalization, multi-head attention, and mixture of exports routing through specialized networks, FFNN 1-N. Medical queries are tokenized while preserving semantic context in multilingual support. Simultaneously, multi-head attention processes pharmaceutical, disease, and clinical concepts. The mixture of exports routing activates experts based on specific medical subdomains, pharmaceutical queries routed through medicine experts, disease queries routed through pathophysiology experts, emergency queries routed through triage experts, etc, while each expert uses feed-forward neural networks with SiLU activation functions, which are optimized for medical subdomain reasoning.

[0015] **FIG. 2 illustrates** the Multi-Layer Architecture, which consists of three layers: Data Layer, AI Intelligence Layer, and Application/Service Layer. In the AI Intelligence Layer, six autonomous AI agents are deployed: Medicine Guidance, Disease Guidance, Medical Report Analyzer,

Geolocation Intelligence, Symptom Inference, Emergency Triage, which are coordinated through Clinical Knowledge Extraction and Adaptive Language Intelligence Engines, while the Application/Service Layer consists of Structured Data Processing, UI Response Generation, Seven Clinical Modules, PDF Email Support, Database, etc.

[0016] Finally, the coordinated autonomous agents process the medical information to provide personalized guidance without any human involvement. The mixture of experts routing optimally allocates resources to the appropriate subdomains, while ensuring clinical consistency. The native language generation maintains precision in medical terms across multiple languages. This model provides democratized access to healthcare services, transcending geographical, economic, and linguistic differences, through continuous intelligent guidance beyond the availability of medical practitioners.

CLAIMS

We claim, Autonomous Multi-Agent Hybrid Healthcare Intelligence System, configured to transform medical information into clinical guidance across multiple languages through specialized autonomous agents without practitioner intervention.

1. The system of claim 1, A self-sufficient multi-agent AI system for guiding healthcare decisions involving a layered three-level structure incorporating multisensory data input, six AI agents designed to work on a transformer

model of neural networks using Mixture-of-Experts (MoE) routing, and application orchestration for context-aware responses in at least twelve language contexts without direct clinician involvement.

2. The system of claim 1, wherein a learning gating mechanism dynamically assigns medical questions to expert networks for pharmaceutical questions, pathophysiology questions, and emergency cases to triage networks while utilizing computational resources according to the complexity of each question without engaging all parameters of the model.

3. The system of claim 1, wherein an Adaptive Language Intelligence Engine creates responses in native languages for medical situations using at least twelve different language platforms by means of language specific expert systems that have been trained on medical literature in their respective languages.

4. The system of claim 1, wherein six asynchronously operating autonomous AI agents performing pharmaceutical interaction analysis, pathophysiology-based differential diagnosis, OCR-based clinical document interpretation, proximity-based geolocation of healthcare facilities, Bayesian probabilistic symptom inference, and standardized emergency triage classification.

5. The system of claim 1, wherein a Clinical Knowledge Extraction Engine autonomously parses pharmaceutical databases and disease taxonomies to extract drug-drug interactions, age-specific contraindications, and evidence-based treatment protocols, representing extracted relationships

as a graph-based knowledge structure enabling multi-hop clinical decision support without real-time database queries.

6. The system of claim 1, a computer-implemented method including receiving heterogenous inputs for medicine, forwarding such inputs to independent AI experts using Mixture-of-Experts models, formulating a response based on probabilistic inference by domain experts, and outputting such results in a user-specified language from among at least twelve languages supported without human involvement.

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ABSTRACT OF THE INVENTION

In this study, we are concerned with Autonomous Multi-Agent Hybrid Healthcare Intelligence System, which offers an intelligent system for transforming diverse medical information into contextualized medical advice through the coordinated efforts of specialized autonomous agents without synchronous medical practitioner intervention. It offers a three-tier system with multimodal medical data ingestion with validation mechanisms, artificial intelligence processing through six specialized autonomous agents with transformer-based neural networks and mixture of experts routing, and application orchestration for offering medical responses in native language formats through multiple linguistic frameworks.

It offers Medicine Guidance, Disease Analysis, Medical Report Interpretation, Geolocation Discovery, Symptom Inference, and Emergency Triage agents, all of which are operating asynchronously in specialized medical subdomains without compromising clinical coherence through structured data processing protocols. It offers a scalable, linguistically diverse, and economically viable system for democratizing preliminary healthcare information across geographical, economic, and linguistic boundaries, facilitating intelligent medical guidance without synchronous medical practitioner intervention.